

Database Connectivity in Python (1)

MARKDOWN

Pre-Requisites

CODE

'''

Database Installed or Accessible

Decide which database you want to connect to, e.g.,:

SQLite – comes built-in with Python (no separate installation needed)

MySQL / MariaDB

PostgreSQL

SQL Server

Oracle

Ensure the database server is running and you have credentials (host, port, username, password, database name).

CODE

'''

Python Code ----- Database Server

Database Connector / Driver

Python cannot talk to databases directly without a connector library.

Examples:

SQLite: sqlite3 (built-in in Python)

MySQL: mysql-connector-python or PyMySQL

PostgreSQL: psycopg2

SQL Server: pyodbc or pymssql

Oracle: cx_Oracle

Install using pip

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Basic Knowledge of SQL

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Step 1: Download MySQL Installer

Go to the official MySQL website: <https://dev.mysql.com/downloads/installer/>

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Step 2: Run the Installer

Launch the downloaded installer.

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Step 3: Install MySQL Server

Installer will guide you to install MySQL Server.

Select latest version unless you have a specific requirement.

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```
####Step 4: Configure MySQL Server
####Choose Standalone MySQL Server.
####Set port (default is 3306).
##Choose Authentication Method:
####Recommended: Use strong password encryption.
####Set root password (remember it, you'll need it for Python).
```

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```
####Step 5: Start MySQL Server
####Ensure MySQL Server service is running.
####You can test using Workbench or command line
```

CODE

```
####Step 7: Install Python Connector
####For Python to connect:
!pip install mysql-connector-python
```

CODE

```
#Go to MySQL Workbench and Type:
#CREATE DATABASE testdb;
#SHOW DATABASES;
#USE testdb

#Creating Tables
'''
-- Table 1: Products
CREATE TABLE Products (
ProductID INT AUTO_INCREMENT PRIMARY KEY,
ProductName VARCHAR(100) NOT NULL,
Category VARCHAR(50),
Price DECIMAL(10, 2), -- Supports prices like 99.99
StockQuantity INT DEFAULT 0
);

-- Table 2: Orders
CREATE TABLE Orders (
OrderID INT AUTO_INCREMENT PRIMARY KEY,
ProductID INT,
CustomerName VARCHAR(100),
OrderDate DATE,
Quantity INT,
-- This links Orders to Products
FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);

'''

##-- Adding Products
'''
INSERT INTO Products (ProductName, Category, Price, StockQuantity)
VALUES
```

```

('Wireless Mouse', 'Electronics', 25.50, 50),
('Mechanical Keyboard', 'Electronics', 75.00, 20),
('Coffee Mug', 'Kitchen', 12.99, 100),
('Desk Lamp', 'Office', 30.00, 15);
'''

#Adding Orders
'''

-- Note: ProductID 1 is the Mouse, ProductID 2 is the Keyboard
INSERT INTO Orders (ProductID, CustomerName, OrderDate, Quantity)
VALUES
(1, 'Alice Smith', '2026-03-20', 2),
(2, 'Bob Jones', '2026-03-21', 1),
(1, 'Charlie Brown', '2026-03-22', 1);
'''

```

CODE

```

import mysql.connector

# 1. Establish the connection
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="admin",
    database="testdb"
)

# 2. Create the cursor
cursor = conn.cursor()

# 3. Execute the command
cursor.execute("SHOW TABLES;")

# 4. FETCH and print the results
for table in cursor.fetchall():
    print(table)
# 5. Clean up
conn.close()

```

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DDL Commands

CODE

```

#1. CREATE
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="admin",
    database="testdb"
)
cursor = conn.cursor()

```

```
# Create a table
cursor.execute("""
CREATE TABLE students (
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(50),
age INT
)
""")
print("Table created successfully!")
conn.close()
#Observe in Workbench
```

CODE

```
##2. ALTER
conn = mysql.connector.connect(
host="localhost",
user="root",
password="admin",
database="testdb"
)
cursor = conn.cursor()

# Add a new column
cursor.execute("ALTER TABLE students ADD COLUMN grade VARCHAR(10)")
print("Column added successfully!")
conn.close()
#Verify from the workbench
```

CODE

```
## ADD THE DATA IN SQL WORKBENCH
'''
INSERT INTO Students (Name, Age, Grade)
VALUES
('Alice Johnson', 20, 'A'),
('Bob Smith', 22, 'B+'),
('Charlie Davis', 21, 'A-'),
('Diana Prince', 23, 'B'),
('Ethan Hunt', 19, 'A');
'''
```

CODE

```
##3.TRUNCATE
conn = mysql.connector.connect(
host="localhost",
user="root",
password="admin",
database="testdb"
)
cursor = conn.cursor()

# Truncate table
```

```
cursor.execute("TRUNCATE TABLE students")
print("Table truncated successfully!")
conn.close()
#Observe from workbench
```

CODE

```
#4. DROP
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="admin",
    database="testdb"
)
cursor = conn.cursor()

# Drop the table
cursor.execute("DROP TABLE students")
print("Table dropped successfully!")
conn.close()
#Observe from workbench
```

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DML

CODE

```
## a) Select all rows
import mysql.connector

with mysql.connector.connect(
    host="localhost",
    user="root",
    password="admin",
    database="testdb"
) as conn:
    cursor = conn.cursor()

    cursor.execute("SELECT * FROM orders")
    rows = cursor.fetchall()

    for row in rows:
        print(row)

#b) Select specific columns
print()
print()
cursor.execute("SELECT CustomerName FROM orders")
rows = cursor.fetchall()
for row in rows:
    print(row)

# SELECT with WHERE filter
print()
```

```
print()
cursor.execute("SELECT CustomerName FROM orders WHERE quantity > 1")
rows = cursor.fetchall()
for row in rows:
    print(row)
```

CODE

```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="admin",
    database="testdb"
)
cursor = conn.cursor()

# Create a table
cursor.execute("""
CREATE TABLE students (
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(50),
age INT
)
""")
print("Table created successfully!")
conn.close()
#Observe in Workbench
```

CODE

```
import mysql.connector

# Connect to MySQL
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="admin",
    database="testdb"
)
cursor = conn.cursor()
```

CODE

```
# Insert a single student
insert_query = "INSERT INTO students (name, age) VALUES (%s, %s)"
student_data = ("Ali", 22)

cursor.execute(insert_query, student_data)
conn.commit()

# Get the auto-generated ID
new_id = cursor.lastrowid
print(f"Inserted student with ID: {new_id}")
```

CODE

```
# Update a student's age
update_query = "UPDATE students SET age = %s WHERE id = %s"
update_data = (23, new_id) # Update the student we just inserted

cursor.execute(update_query, update_data)
conn.commit()

print(f"Updated student ID {new_id} successfully!")
```

CODE

```
cursor.execute("UPDATE students SET name = %s, age = %s WHERE id = %s", ("Shoaib", 24,
new_id))
conn.commit()
```

CODE

```
# Delete the student
delete_query = "DELETE FROM students WHERE id = %s"
delete_data = (new_id,)

cursor.execute(delete_query, delete_data)
conn.commit()

print(f"Deleted student ID {new_id} successfully!")
```

CODE

```
cursor.close()
conn.close()
```